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CO₂ Storage Risk Assessment: Leveraging Machine Learning Proxy Modeling for Predicting Fracture Growth and Leakage

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Abstract

The growing deployment of carbon capture and storage (CCS) requires robust site screening and risk assessment tools to ensure safe and efficient CO₂ injection. Traditional reservoir simulations, while highly accurate, are computationally expensive and impractical for real-time risk evaluation. Existing proxy models are unable to associate real time subsurface conditions with the extent of the fracture created, limiting their ability to inform real time operational decisions. This study aims to develop a machine learning (ML)-assisted proxy model that can predict critical subsurface integrity issues by linking subsurface conditions to fracture likelihood, enabling proactive evaluation of storage risks before injection and improving risk mitigation strategies through real-time monitoring data.

A high-fidelity numerical simulator was used to generate a dataset of 150 simulation cases, covering diverse geological and operational conditions derived from actual CCS projects. Latin Hypercube Sampling (LHS) was employed to ensure comprehensive parameter coverage while maintaining data sparsity. The ML proxy model was trained using a neural network algorithm, testing different combinations of hyper-parameters until an optimum combination was found that ensured prediction accuracy while avoiding overfitting. The model was used to predict the occurrence of fractures. Model reliability was evaluated using key performance metrics such as Mean Absolute Error (MAE), R² scores, Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE). Additionally, the developed model is tested with real field datasets to validate its applicability in practical storage scenarios.

The ML-assisted proxy model demonstrated a high degree of accuracy compared to full-physics reservoir simulations, significantly reducing computational costs while maintaining predictive reliability. Results showed that the model successfully identified scenarios where fractures happen and could lead to CO₂ leakage, quantifying fracture occurrence to inform corrective actions, such as reducing injection rates or increasing production from a pump well. Model also shows what well monitoring data (temperature and pressure) could look like in real-time, making it a valuable tool for offering dynamic insights into subsurface behavior. This study presents a scalable framework for site selection and operational decision-making. This work introduces a novel approach to evaluating subsurface CO₂ storage integrity risks with high accuracy and real-time adaptability. Unlike existing methodologies, it accurately predicts fracture

occurrence allowing for proactive risk assessment before injection and operational adjustments during storage. The opportunity to integrate real-time monitoring data to this model transforms conventional CCS risk management, making large-scale CO₂ storage safer and more predictable. The broader impact extends beyond CCS, offering a novel framework applicable to underground hydrogen storage and geothermal reservoir integrity assessments.

Introduction

Rising atmospheric carbon dioxide (CO₂) concentrations, currently exceeding 420 ppm compared to pre-industrial levels of approximately 280 ppm, have been a significant driver of anthropogenic climate change (IPCC, 2005; Hashimoto & Hashimoto, 2019). Among the most promising approaches to mitigating anthropogenic CO₂ emissions is geological carbon sequestration, a process in which CO₂ is injected into deep subsurface formations such as saline aquifers, depleted oil and gas reservoirs, and reactive rock formations like basalts (Massarweh & Abushaikh, 2024). Through a combination of trapping mechanisms—structural, residual, solubility, and mineral trapping—geological sequestration offers long-term, secure containment of CO₂ (Benson & Cole, 2008). However, the success of these operations depends on a detailed understanding of subsurface behavior during and after injection. Predicting how CO₂ migrates, interacts with formation fluids and rocks, and affects reservoir pressure and geomechanics is crucial for ensuring containment integrity and avoiding induced hazards such as fracture propagation or fault reactivation. Due to the challenges in conducting large-scale experimental studies under deep subsurface conditions, numerical simulation has become an indispensable tool for assessing and optimizing CO₂ storage operations (Dai et al., 2020; Sori et al., 2024). In particular, coupled thermo-hydro-mechanical (THM) simulations allow for simultaneous modeling of fluid flow, heat transfer, geomechanical stress evolution, and fracture behavior within the reservoir (Rutqvist, 2012). These models support robust scenario analysis but are typically used in pre-operational planning or periodic updates, rather than in real-time decision-making due to their computational intensity and limited capacity to assimilate real-time data.

To address this limitation, digital twins are emerging as a critical innovation in subsurface engineering. A digital twin is a dynamic, data-driven virtual replica of a physical asset or system that integrates real-time sensor data, simulation models, and analytics for continuous monitoring and control (Khalili & Ahmadi, 2023). In the context of geological CO₂ sequestration, a digital twin enables live tracking of bottom-hole pressure (BHP), bottom-hole temperature (BHT), and geomechanical responses, while also offering operational insights to support risk mitigation and process optimization (Balachandran & Padmanabhan, 2023). Despite its potential, the application of digital twins to subsurface flow and CO₂ injection remains relatively underexplored.

Related Studies

Several recent studies have attempted to bridge this gap. For example, Stepien et al. (2022) utilized a generative adversarial network (GAN) to predict CO₂ plume migration using sparse well data, dramatically reducing computational time. However, their model focuses solely on plume evolution and does not account for mechanical risks such as fracture propagation. Similarly, Stepien et al. (2022) developed a GAN-based digital twin for plume prediction; however, it lacked real-time operational feedback, providing limited guidance for field adjustments. Yang et al. (2024) introduced a digital twin for underground gas storage (UGS) that incorporates THM simulations and machine learning diagnostics; however, it falls short of delivering actionable recommendations in real-time.

Table 1—Review of related subsurface digital twin works

Reference	Problem statement	Methodology	Limitation
Yang et al., 2024	Real-time risk identification, flow assurance, and optimized control in UGS operations are hindered by limitations in data integration and predictive capabilities.	A digital twin was developed, integrating THM simulation, ML-assisted diagnostics, and real-time sensor data into a unified, dashboard-accessible platform.	Real-time diagnostics, but no explicit suggestions
Stepien et al., 2022	Traditional numerical simulations of CO ₂ plume evolution are too computationally intensive for real-time monitoring.	A continuous conditional generative adversarial network (CCGAN) is trained using well data to rapidly predict CO ₂ saturation fields.	Focuses only on CO ₂ plume evolution, not on mechanical risks like fracturing.

Addressing the limitations of previous approaches, such as the lack of explicit recommendations and a sole focus on CO₂ plume evolution, this study introduces a comprehensive proxy framework that serves as a precursor to a digital twin (without front-end visualization), specifically tailored for geological CO₂ sequestration. This framework emphasizes the backend components, including anomaly detection, proxy modeling, and recommendation systems. It integrates high-fidelity thermo-hydro-mechanical (THM) simulations with machine learning-based proxy models, SHAP-enabled explainable AI, real-time anomaly detection, and a live recommendation engine. Through a Message Queuing Telemetry Transport (MQTT)-based communication layer, the system continuously transfers bottomhole pressure (BHP) and bottomhole temperature (BHT) to identify early fracture risk indicators, enabling the early detection of potential mechanical failures and autonomously providing actionable mitigation strategies, such as optimizing injection rates or initiating shut-in protocols.

Also, recent studies have focused on developing risk mitigation strategies and management plans in advance of project execution. For example, [Badejo et al. \(2024\)](#) identified key risks and proposed an environmental risk monitoring and management framework for a major CCS project in the UK, addressing the pre-injection, injection, and post-injection phases, along with associated mitigation measures. However, these approaches do not incorporate real-time monitoring. The novelty of our work lies in the integration of real-time risk monitoring during the operational phase, enabling more responsive and adaptive risk management.

The primary objective of this project is to demonstrate that a fully integrated proxy framework can improve the safety and efficiency of CO₂ injection by converting predictive insights into real-time operational control. By embedding explainable AI, it ensures transparency and scientific validity in decision-making, while providing a scalable methodology to track, predict, and respond to subsurface changes as an intelligent control layer that bridges simulation and field operations. Also, in this work, the emphasis is placed specifically on the backend architecture because, in a digital twin, it serves as the computational and decision-making hub, executing high-fidelity simulations, AI models, and real-time analytics to detect anomalies and generate recommendations, thereby enabling the predictive, diagnostic, and prescriptive capabilities critical for accurate, timely, and automated operations.

Methodology

This project employed a multi-stage workflow that integrates high-fidelity numerical simulations, machine learning, anomaly detection, and a recommendation system to develop an intelligent proxy system for CO₂ sequestration monitoring and control. The methodology incorporates a fully coupled thermo-hydro-mechanical (THM) simulator to evaluate various operational scenarios, generating a comprehensive dataset for machine learning-based proxy models. These proxy models enable rapid sensitivity analysis, anomaly detection, and scenario forecasting. A central innovation of this study is the deployment of a real-time recommendation engine capable of detecting the onset of fractures by monitoring bottom-hole pressure (BHP), bottom-hole temperature (BHT), and geomechanical indicators. Upon identifying potential fracture

risks, the system autonomously recommends operational adjustments, such as modifying the injection rate or temporarily shutting in, to mitigate damage and maintain storage integrity. The detailed workflow is as discussed in the sections that follow. The overall methodology is illustrated in Fig. 1.

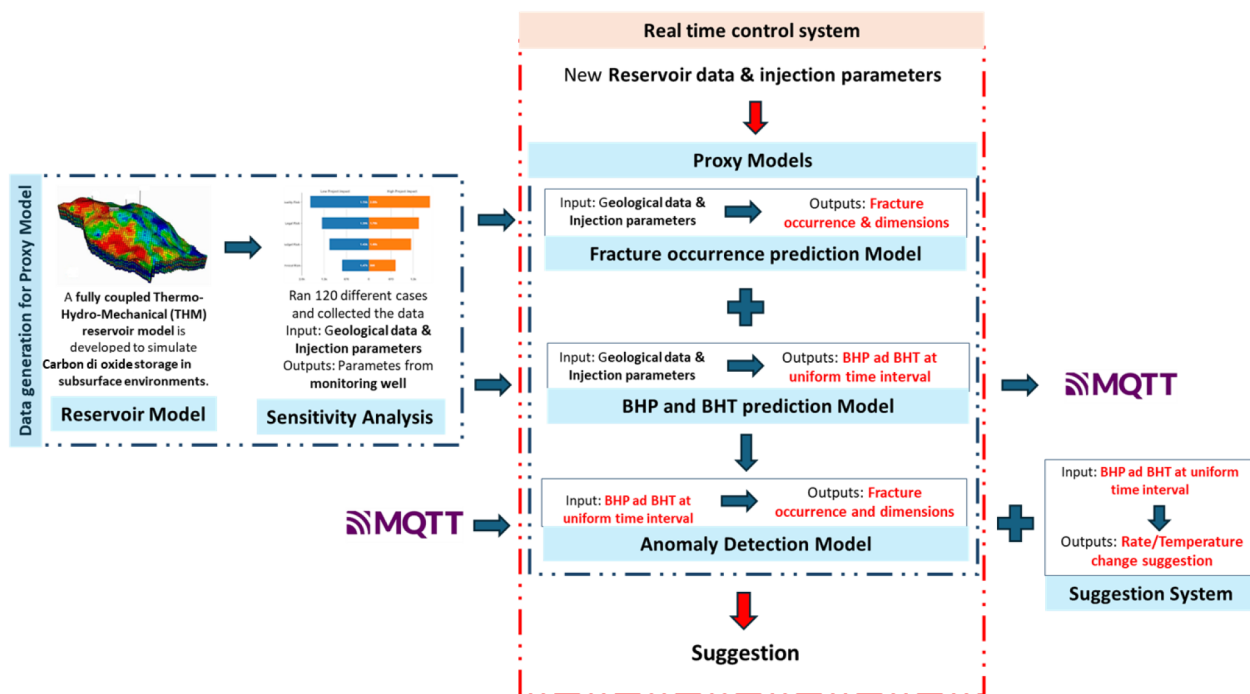


Figure 1—Methodology flowchart

Fully Coupled Thermo-Hydro-Mechanical (THM) Simulation

A fully coupled Thermo-Hydro-Mechanical (THM) reservoir simulator was developed to replicate subsurface CO₂ injection and evaluate fracture risk. The simulator captures fluid flow, heat transfer, and geomechanical deformation to realistically represent reservoir behavior. The simulation framework was adapted from McClure et al. (2018) and customized to analyze fracture initiation over a one-year continuous injection period.

Sensitivity Analysis and Data Generation

Latin Hypercube Sampling (LHS) was employed to systematically explore the input parameter space. A total of 150 simulation cases were generated, varying key geological and operational parameters, such as reservoir depth, permeability, Young's modulus, and injection pressure. These variations ensure broad coverage of realistic subsurface conditions. The full range of input parameters is summarized in Table 2. Synthetic data from 150 simulations, varying geological and operational parameters, were used to generate bottom-hole pressure (BHP), bottom-hole temperature (BHT), and fracture occurrence. These results from the fully coupled THM simulations formed the training dataset for subsequent machine learning tasks. The key outputs included fracture occurrence in a binary form, and time-series BHP and BHT profiles.

Table 2—Input ranges for data generation

Parameters	Min	Base	Max
Reservoir Top (m)	600	1500	4000
Reservoir Thickness (m)	10	100	300
Reservoir Permeability (mD)	10	100	5000
Kv/Kh	0.01	0.1	0.5

Parameters	Min	Base	Max
Porosity	0.05	0.2	0.35
Minimum Horizontal Stress (MPa/km)	10	15	30
Young's Modulus (GPa)	7	15	50
Uniaxial Compressive Strength (MPa)	10	15	50
Perforation interval (% of bed thickness)	10	50	100

Feature Engineering and Dimensionality Reduction

From an initial set of 32 features, including geological boundaries, directional permeability components, stress states, and injection conditions, feature selection was performed using Spearman correlation analysis. This process reduced the feature set to 9 influential input variables, improving model efficiency while preserving predictive accuracy.

Proxy Modeling

The structured dataset was organized into three primary tables: (1) an input parameter table, (2) an output table containing fracture-related parameters, and (3) a time-series table of BHP and BHT values. Based on these, three neural network-based proxy models were developed to approximate complex THM processes and support real-time evaluation.

The models are as follows:

- **Fracture Classification Model:** Predicts the occurrence (binary) of fractures using geological and operational inputs.
- **Thermal-Hydraulic Prediction Model:** Forecasts the temporal evolution of BHP and BHT during CO₂ injection, capturing the system's thermal and hydraulic responses.
- **Anomaly Detection Model:** Detects deviations in BHP/BHT trends indicative of fracture onset, leveraging pattern recognition from simulation-derived signatures.

Real-Time Streaming with MQTT

To emulate real-time field conditions, predicted BHP and BHT values from the thermal-hydraulic model were streamed using MQTT (Message Queuing Telemetry Transport). This protocol simulates the transmission of live sensor data from the reservoir, enabling continuous monitoring and integration with the anomaly detection module.

SHAP-Based Anomaly Diagnosis and Recommendation System

To ensure interpretability and guide operational decisions, SHAP (SHapley Additive exPlanations) values were used to explain predictions from the anomaly detection model. SHAP analysis identifies whether pressure or temperature is the dominant contributor to fracture risk. Based on this attribution, the system provides actionable recommendations: **"Reduce the injection rate"** if pressure is the dominant driver. **"Lower the injection temperature"** if thermal effects are the primary cause. This explainable AI-driven mechanism adds a decision-support layer to the framework.

Real-Time Visualization with Unreal Engine

A real-time 3D visualization platform was developed using Unreal Engine to enhance the interpretability of subsurface processes. The interface visualizes CO₂ plume migration, fracture evolution, and BHP/BHT dynamics in an interactive environment. Live data streams from the proxy models, transmitted via MQTT,

are rendered in real-time, enabling operators and stakeholders to monitor reservoir behavior and respond to anomalies efficiently.

Results

The developed digital twin framework was evaluated using a newly generated dataset. New input scenarios were passed through the trained proxy models, and the resulting outputs were compared against the anomaly dataset. If the output signature matched a known anomaly pattern, it was flagged as a fracture risk. This approach enables rapid, real-time detection of abnormal behavior during the injection phase, thereby facilitating proactive operational intervention.

Simulation and Sensitivity Analysis Results

The fully coupled THM simulations demonstrated the sensitivity of fracture formation to variations in both geological and operational parameters. A one-year continuous CO₂ injection period was modeled to assess the likelihood of fracture initiation under a wide range of subsurface conditions. These simulations provided insights into the mechanical response of the reservoir and informed the training of the proxy models.

Table 3 summarizes the outputs from the THM model, including fracture surface area, length, height, and plume diameter. The simulation results captured multiple aspects of reservoir response during CO₂ injection. These included the evolution of fracture dimensions, such as length, height, and surface area, which serve as key indicators of potential mechanical failure. Also, the model recorded plume size to assess CO₂ migration, along with time-series data for bottom-hole pressure (BHP) and bottom-hole temperature (BHT) to monitor thermal and hydraulic behavior over time.

Table 3—Output parameters showing the fracture dimensions

Max Fracture Surface Area (m ²)	Max Fracture Length (m)	Max Fracture Height (m)	Gas Plume Diameter (m)	Fracture occurrence
0	0	0	1282.464	Yes
2877.5873	55.351084	54.653766	849.98	No

Feature Importance and Interpretability

The SHAP analysis identified reservoir thickness, permeability, and minimum horizontal stress as the dominant predictors of fracture occurrence (Fig. 2). These findings align well with established geomechanical theory, reinforcing the credibility of the model. Reservoir thickness directly influences the vertical extent available for fracture growth. Higher permeability enhances fluid mobility, which affects pressure buildup and increases the likelihood of fracture initiation. Minimum horizontal stress (Shmin) plays a pivotal role in determining both the pressure required for fracture initiation and the direction of fracture propagation, as fractures tend to propagate perpendicular to the least principal stress (Zoback, 2010). Thus, the strong influence of these parameters on fracture dimensions is consistent with known geomechanical principles. The crucial input parameters of importance are summarized in Table 4.

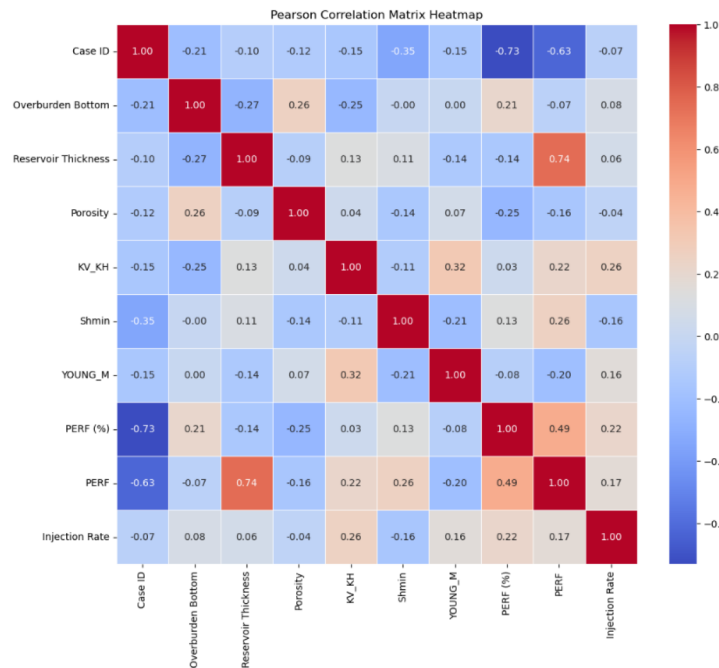


Figure 2—Pearson correlation plot after feature reduction.

Table 4—Input parameters after feature importance

Reservoir Thickness	Perm	Kv_Kh	Zperm	Shmin	Maximum Injection Pressure	Young_M	Perf (%)	Perf
19.7095	223.30	0.150396	3.36E-14	26.1822	50.16747778	8.4102	65.917	12.99191

Proxy Model Performance

Three proxy models were developed as part of the digital twin framework. The fracture classification model demonstrated high accuracy in distinguishing between fracture-prone and safe injection scenarios. Fig. 3 presents a comparison between predicted and simulated fracture feature values for a representative case. The BHP/BHT prediction model effectively reproduced the thermal and hydraulic trends observed in the full-physics simulations. The average relative error remained below 15%, demonstrating strong predictive fidelity. Table 5 summarizes the predicted bottom-hole pressure and temperature profiles at uniform time intervals.

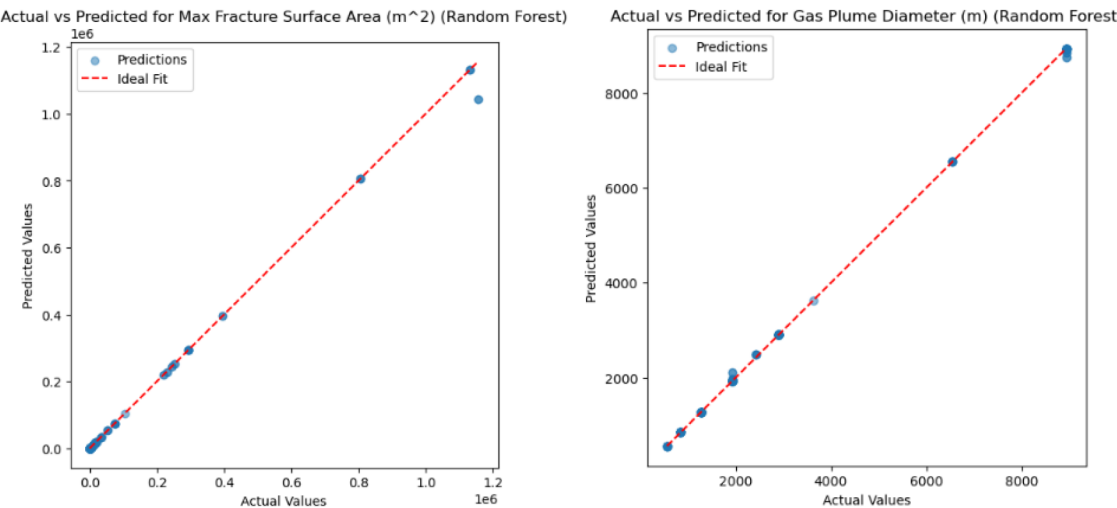


Figure 3—Comparison of predicted vs. simulated fracture feature values for a representative case

Table 5—Predicted bottom-hole temperature and pressure over time

Step	Time	BHT	BHP
0	1.0e-03	14.43	65.81
1	7.9e+06	17.83	65.95
2	1.5e+07	18.56	65.95
3	2.3e+07	18.83	65.95
4	3.1e+07	19.03	65.95

For classification simplicity, the initial phase of the study treated the occurrence of fracturing as a binary outcome indicating either the presence ("Yes") or absence ("No") of fracture activity. To simulate real-time field conditions, the predicted bottom-hole pressure (BHP) and bottom-hole temperature (BHT) values generated by the thermal-hydraulic proxy model were streamed using the Message Queuing Telemetry Transport (MQTT) protocol. A representative sample of time-series data transmitted via MQTT, including predicted BHP and BHT values, is presented in Table 5.

Real-Time Monitoring and Recommendation Results

During real-time emulation, the anomaly detection model reliably flagged pressure-induced fractures and issued appropriate operational recommendations based on SHAP interpretability. Table 6 presents example outputs from the system, including BHP/BHT values, anomaly status, the dominant contributing factor, and the suggested operational response.

Table 6—Anomaly detection and real-time operational recommendations

BHP (MPa)	BHT (°C)	Anomaly Detected	Dominant Feature	Suggested Action
65.95	18.83	Yes	Pressure	Reduce Injection Rate
65.80	14.43	No	--	Continue Operation

Visualization Outcome

The 3D visualization interface, developed using Unreal Engine, successfully rendered dynamic processes, such as CO₂ injection and plume migration, in real-time. The interface supports toggling between simulation cases and includes visual alerts upon detection of threshold breaches, such as the onset of fracture. This

feature enhances system usability for both technical users and non-expert stakeholders by providing an intuitive, interactive understanding of subsurface behavior.

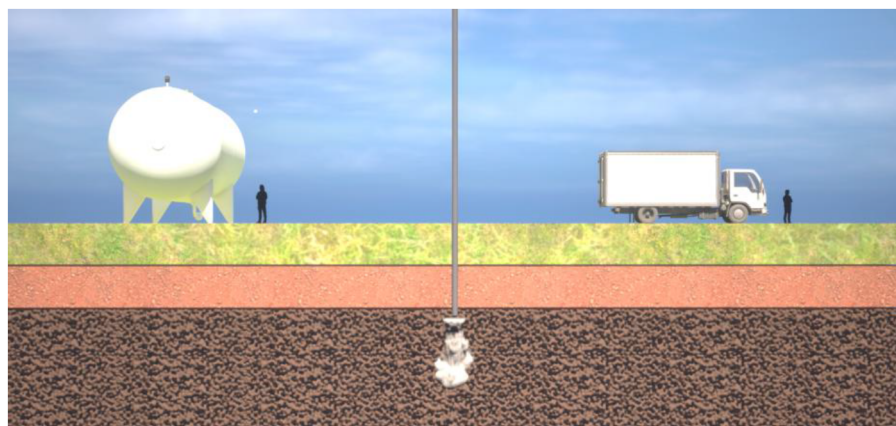


Figure 4—Visualization model

Discussion

The results of this study demonstrate the effectiveness and versatility of a digital twin framework for real-time monitoring and control of CO₂ sequestration operations. By combining fully coupled Thermo-Hydro-Mechanical (THM) simulations with machine learning-based proxy models, the system delivers both predictive performance and operational insight. The framework fulfills the project's primary objectives: accurately identifying fracture risks, providing real-time pressure and temperature predictions, and offering actionable recommendations through explainable AI. The proxy model discussions are summarized below:

- The fracture classification model showed strong agreement with the outcomes of high-fidelity THM simulations, indicating its reliability in identifying mechanically unstable injection conditions. This is particularly valuable in field-scale applications, where running full THM simulations for every scenario is computationally infeasible. By replacing these with rapid proxy-based approximations, operators can receive real-time feedback on injection safety under evolving reservoir conditions.
- The anomaly detection module represents a key innovation of the workflow. Through continuous comparison of predicted BHP and BHT values with known anomalous signatures, the system identifies early signs of fracture initiation. SHAP analysis enhances trust in these predictions by attributing risk to specific physical drivers, such as high injection pressures or excessive thermal contrasts. This capability offers a clear advantage over black-box models, which provide limited interpretability.
- The real-time emulation enabled by MQTT effectively replicates live sensor feedback, showcasing the feasibility of deploying this framework in operational settings. Streaming predicted BHP and BHT values allows the system to remain continuously informed and responsive, thereby supporting near-instantaneous adjustments to injection parameters.
- The Unreal Engine-based 3D visualization further extends the utility of the digital twin. By converting complex simulation results into immersive, intuitive graphics, it supports cross-disciplinary communication and stakeholder engagement. This is particularly critical in CCS projects, where public understanding and regulatory approval often hinge on clear communication of subsurface risks and performance.

Despite its strengths, the study has certain limitations. The simulations are based on synthetic data and assume idealized geological conditions. While Latin Hypercube Sampling ensures coverage of a broad

parameter space, site-specific variability may introduce unforeseen complexities. Additionally, while the neural network models perform well on validation data, their robustness in the presence of real-world noise and sensor error remains to be tested. Future studies should include integration with real field data, uncertainty quantification, and exploration of edge deployment strategies for field-ready implementations. Moreover, expanding the system to incorporate reactive transport and geochemical interactions would provide a more holistic digital twin for CCS.

Conclusion

This study presents a robust digital twin architecture for geological carbon sequestration, integrating THM simulations, machine learning-based proxy modeling, real-time anomaly detection, and 3D visualization. By replicating reservoir behavior under various geological and operational scenarios, the digital twin successfully predicted trends in fracture risk, pressure, and temperature, and provided interpretable recommendations for safe injection strategies.

Key contributions include:

- A high-accuracy fracture classification model linked to THM outputs
- Real-time emulation of BHP and BHT using MQTT communication
- Anomaly detection powered by SHAP-based interpretability for operational guidance
- A fully interactive Unreal Engine visualization interface for stakeholder engagement

The framework bridges the gap between computational modeling and operational control, enabling real-time decision-making rooted in both physics and data. It lays the foundation for more intelligent, adaptive, and risk-aware CO₂ storage operations. The research outcomes have practical implications for subsurface monitoring, injection optimization, and automated reservoir management in carbon storage projects. This approach can also be extended to other subsurface applications such as hydrogen storage, geothermal systems, and underground gas storage (UGS), supporting broader energy transition goals.

Future Work

Building on the promising results of this study, several directions for future research and system enhancement are identified:

1. **Field-Scale Validation:** Incorporating real-time data from pilot or commercial-scale CO₂ sequestration sites is essential to validate the robustness of proxy models under real-world uncertainty and noise. This includes sensor calibration, handling missing data, and integrating site-specific geological models.
2. **Geochemical and Reactive Transport Coupling:** Current models focus on THM processes. Future work should expand to incorporate reactive transport and CO₂-mineral interactions, enabling more complete risk assessments, especially in reactive formations such as basalt or ultramafic rocks.
3. **Uncertainty Quantification and Risk Scoring:** Implementing probabilistic methods (e.g., Monte Carlo simulations or Bayesian neural networks) would allow the system to output confidence levels alongside predictions, supporting more informed risk-based decisions.
4. **Integration with Edge Devices:** Developing lightweight model versions deployable on edge computing platforms would allow onsite monitoring without reliance on centralized infrastructure, particularly useful in remote injection sites.

5. **User Interface and Alert Customization:** Adding user-configurable thresholds, alert levels, and control logic would allow operators to tailor the digital twin to specific project objectives and regulatory frameworks.
6. **Extension to Multi-Well Systems:** While the current workflow focuses on single-well injection, future work should include multi-well configurations and regional-scale plume interaction to support full-site CCS management.
7. **Application to Other Subsurface Domains:** The digital twin architecture can be adapted for use in geothermal operations, hydrogen storage, or waste injection, providing a unified monitoring platform for various low-carbon energy technologies.

Addressing these areas will result in a digital twin framework that can evolve into a versatile, field-ready solution for intelligent subsurface energy system management.

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